The Galactic tidal field in HENDOS

HErmite N-body Dynamics Of Systems

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1 Overview

The cluster is assumed to move on a circular orbit in an axisymmetric Galaxy. The integration is carried out in a frame centred on the cluster's guiding centre and rotating with the orbit, and the Galactic potential is expanded to second order about the guiding centre (Hill's approximation; e.g. Chandrasekhar 1942; Heggie & Hut 2003; Renaud, Gieles & Boily 2011). The external field then enters the equations of motion as a constant, symmetric *tidal tensor* T plus the Coriolis force. Because T is constant in the rotating frame, the Jacobi energy is conserved and can be used to monitor the accuracy of the integration in the same way as the energy of an isolated cluster.

Three options are available through the parameter `tidal` in the `&tides` section of the parameter file:

- `'none'`: an isolated cluster ($\mathsf{T} = 0$, $\Omega = 0$);
- `'solar'`: the solar-neighbourhood field, built from the Oort constants and the local mass density (Section 4);
- `'tensor'`: any user-supplied tensor T and frame rotation rate Ω.

2 Coordinate frame

The frame rotates with angular velocity $\boldsymbol{\Omega} = \Omega\,\hat{\mathbf{z}}$ about the cluster centre, with

- x pointing away from the Galactic centre,
- y pointing in the direction of Galactic rotation,
- z completing a right-handed set, i.e. along the Galaxy's spin (towards the south Galactic pole).

Initial positions and velocities are read, and snapshots are written, in this frame. Velocities are therefore velocities *relative to the rotating frame*. A cluster that is non-rotating in an inertial frame appears to rotate at $-\Omega$ in this frame, but for the solar value ($\Omega \simeq 27\,\mathrm{km\,s^{-1}\,kpc^{-1}}$) this amounts to only $\sim 0.03\,\mathrm{km\,s^{-1}}$ per pc of radius, which is negligible compared with typical velocity dispersions.

3 Equations of motion

For star i at position $\mathbf{x}_i$ with velocity $\mathbf{v}_i$ in the rotating frame,

$$\ddot{\mathbf{x}}_i = \mathbf{a}_{\mathrm{grav},i} + \mathsf{T}\mathbf{x}_i - 2\boldsymbol{\Omega} \times \mathbf{v}_i, \tag{1}$$

where $\mathbf{a}_{\mathrm{grav},i}$ is the gravitational acceleration from the other stars. The term $\mathsf{T}\mathbf{x}$ is the tidal acceleration including the centrifugal term, and $-2\boldsymbol{\Omega}\times\mathbf{v}$ is the Coriolis acceleration. In components, with $\boldsymbol{\Omega}=\Omega\hat{\mathbf{z}}$,

$$-2\boldsymbol{\Omega}\times\mathbf{v} = 2\Omega\,(v_y,\,-v_x,\,0). \tag{2}$$

The effective tensor is

$$\mathsf{T} = \mathsf{T}_{\mathrm{G}} + \Omega^2\,\mathrm{diag}(1,1,0), \qquad (\mathsf{T}_{\mathrm{G}})_{kl} = -\left.\frac{\partial^2\Phi_{\mathrm{G}}}{\partial x_k\,\partial x_l}\right|_{\mathrm{gc}}, \tag{3}$$

where Φ_{G} is the Galactic potential and the derivatives are taken at the guiding centre. The second term is the centrifugal contribution of the rotating frame. T must be symmetric; if a non-symmetric tensor is supplied, the code warns and uses its symmetric part $(\mathsf{T}+\mathsf{T}^{\mathsf{T}})/2$.

4 The solar-neighbourhood field

For an axisymmetric potential with circular speed $v_c(R)$ the angular velocity of the frame and the derivatives of Φ_{G} at the guiding centre can be written in terms of the Oort constants $A=-\frac{1}{2}R\,\mathrm{d}\Omega/\mathrm{d}R$ and $B=A-\Omega$:

$$\Omega = A - B, \tag{4}$$

$$\frac{\partial^2\Phi_{\mathrm{G}}}{\partial R^2} = \Omega^2 - 4A\Omega, \tag{5}$$

$$\frac{1}{R}\frac{\partial\Phi_{\mathrm{G}}}{\partial R} = \Omega^2, \tag{6}$$

$$\frac{\partial^2\Phi_{\mathrm{G}}}{\partial z^2} \equiv \nu^2 = 4\pi G\rho_0 + 2\left(A^2-B^2\right), \tag{7}$$

where the last line follows from Poisson's equation in cylindrical coordinates and ρ_0 is the total (stars, gas and dark matter) mass density at the guiding centre. Adding the centrifugal term gives

$$\mathsf{T} = \mathrm{diag}\left(4A(A-B),\;0,\;-\left[4\pi G\rho_0 + 2\left(A^2-B^2\right)\right]\right). \tag{8}$$

The equations of motion then reduce to the familiar Hill equations

$$\ddot{x} = a_{\mathrm{grav},x} + 4A\Omega\,x + 2\Omega\,\dot{y}, \tag{9}$$

$$\ddot{y} = a_{\mathrm{grav},y} - 2\Omega\,\dot{x}, \tag{10}$$

$$\ddot{z} = a_{\mathrm{grav},z} - \nu^2 z. \tag{11}$$

The default values are given in Table 1. They can be changed in the parameter file.

Quantity	Value	Source
A	$15.3\,\mathrm{km\,s^{-1}\,kpc^{-1}}$	Bovy (2017)
B	$-11.9\,\mathrm{km\,s^{-1}\,kpc^{-1}}$	Bovy (2017)
ρ_0	$0.097\,\mathrm{M_\odot\,pc^{-3}}$	McKee, Parravano & Hollenbach (2015)
$\Omega = A-B$	$27.2\,\mathrm{km\,s^{-1}\,kpc^{-1}}$	
$T_{xx} = 4A\Omega$	$1664.6\,(\mathrm{km\,s^{-1}\,kpc^{-1}})^2$	
$T_{zz} = -\nu^2$	$-5427.5\,(\mathrm{km\,s^{-1}\,kpc^{-1}})^2$	

Table 1: Default solar-neighbourhood parameters and the resulting tidal field.

4.1 Jacobi radius

For a cluster of mass M the Jacobi (tidal) radius along the x-axis is

$$r_{\rm J} = \left(\frac{GM}{T_{xx}}\right)^{1/3} = \left(\frac{GM}{4A\Omega}\right)^{1/3}, \tag{12}$$

which the code prints at the start of any run with $T_{xx} > 0$. For $M = 1000\,{\rm M}_\odot$ in the solar neighbourhood $r_{\rm J} \simeq 13.7\,{\rm pc}$.

5 Energy

The Coriolis force does no work and T is constant and symmetric, so the tidal acceleration derives from the potential

$$\Phi_{\rm tid}(\mathbf{x}) = -\tfrac{1}{2}\,\mathbf{x}^{\mathsf{T}}\,\mathsf{T}\,\mathbf{x}. \tag{13}$$

The conserved quantity is the Jacobi energy

$$E_{\rm J} = \sum_i \tfrac{1}{2} m_i |\mathbf{v}_i|^2 \;-\; \sum_{i<j} \frac{G m_i m_j}{\left(r_{ij}^2 + \epsilon^2\right)^{1/2}} \;+\; \sum_i m_i\,\Phi_{\rm tid}(\mathbf{x}_i), \tag{14}$$

with velocities measured in the rotating frame. When a tide is switched on, the energy and relative energy error printed by the code are those of $E_{\rm J}$. The virial ratio printed is still $-K/W$ using the gravitational potential energy W alone.

6 Implementation in the Hermite scheme

6.1 Acceleration and jerk

The fourth-order Hermite scheme needs both the acceleration and its time derivative (the jerk) for each star. Differentiating equation (1) gives the tidal contribution to the jerk,

$$\mathbf{j}_{\rm tid} = \mathsf{T}\mathbf{v} - 2\boldsymbol{\Omega} \times \mathbf{a}, \tag{15}$$

where $\mathbf{a}$ is the *total* acceleration (gravitational plus tidal plus Coriolis). In the force routine the gravitational sum over all other stars is done first; the tidal and Coriolis accelerations are then added; and only then are the tidal jerk terms added, so that they use the complete acceleration:

```
ax = ax + T11*x + T12*y + T13*z + 2*omega*vy
ay = ay + T21*x + T22*y + T23*z - 2*omega*vx
az = az + T31*x + T32*y + T33*z
jx = jx + T11*vx + T12*vy + T13*vz + 2*omega*ay
jy = jy + T21*vx + T22*vy + T23*vz - 2*omega*ax
jz = jz + T31*vx + T32*vy + T33*vz
```

The cost is a few operations per active star per step, negligible compared with the $O(N)$ gravitational sum.

6.2 Predictor order

In the standard Hermite scheme positions are predicted to third order and velocities to second order,

$$\mathbf{x}_{\rm p} = \mathbf{x}_0 + \mathbf{v}_0\Delta t + \tfrac{1}{2}\mathbf{a}_0\Delta t^2 + \tfrac{1}{6}\mathbf{j}_0\Delta t^3, \tag{16}$$

$$\mathbf{v}_{\rm p} = \mathbf{v}_0 + \mathbf{a}_0\Delta t + \tfrac{1}{2}\mathbf{j}_0\Delta t^2. \tag{17}$$

For purely gravitational forces, which depend only on positions, this is sufficient for fourth-order accuracy. The Coriolis force, however, depends on velocity, and an $O(\Delta t^3)$ error in $\mathbf{v}_\mathrm{p}$ then gives an $O(\Delta t^3)$ error in the force evaluated at the predicted state. This reduces the whole scheme to third order. Tests confirmed this: with a strong tide the energy error fell by a factor of ≈ 8, not 16, for each halving of the timestep.

The code therefore raises the order of the prediction used for the force evaluation by adding the second and third derivatives of the acceleration, $\mathbf{a}^{(2)}$ and $\mathbf{a}^{(3)}$, from the end of each star's previous step:

$$\mathbf{x}_\mathrm{p} \leftarrow \mathbf{x}_\mathrm{p} + \tfrac{1}{24}\mathbf{a}^{(2)}\Delta t^4 + \tfrac{1}{120}\mathbf{a}^{(3)}\Delta t^5, \tag{18}$$

$$\mathbf{v}_\mathrm{p} \leftarrow \mathbf{v}_\mathrm{p} + \tfrac{1}{6}\mathbf{a}^{(2)}\Delta t^3 + \tfrac{1}{24}\mathbf{a}^{(3)}\Delta t^4. \tag{19}$$

Here $\mathbf{a}^{(2)} = \mathbf{f}_2 + \mathbf{f}_3\Delta t_\mathrm{prev}$ and $\mathbf{a}^{(3)} = \mathbf{f}_3$ come from the previous corrector step (arrays `f2e`, `f3e`); they are zero on the first step. The corrector itself is unchanged and still starts from the third-order prediction, because it adds the $\mathbf{f}_2$ and $\mathbf{f}_3$ terms itself; using the higher-order prediction there would count those terms twice. The improved prediction is also used for the non-active stars, which helps the accuracy of block timesteps in general.

6.3 Timesteps

No special timestep criterion is needed for the tide. The Aarseth criterion uses the total acceleration and its derivatives, so stars far from the cluster, whose motion is dominated by the tide, automatically take steps set by the tidal frequencies. The largest step any star can take is the output interval `tout`.

6.4 Units

Internally the code uses $G = 1$, $\mathrm{M}_\odot$ and pc, giving a velocity unit of $0.06558\,\mathrm{km\,s^{-1}}$ and a time unit of $14.91\,\mathrm{Myr}$. The tidal parameters are read in $\mathrm{km\,s^{-1}\,kpc^{-1}}$ (rates) and $(\mathrm{km\,s^{-1}\,kpc^{-1}})^2$ (tensor components), and converted with

$$1\,\mathrm{km\,s^{-1}\,kpc^{-1}} = \frac{1}{1000\,v_\mathrm{unit}}\ \text{internal units}, \tag{20}$$

where v_unit is in $\mathrm{km\,s^{-1}}$. Because the internal mass and length units are $\mathrm{M}_\odot$ and pc with $G = 1$, the density term $4\pi G\rho_0$ is simply $4\pi\rho_0$ with ρ_0 in $\mathrm{M_\odot\,pc^{-3}}$. The user never sees internal units: the tensor and Ω are echoed in $\mathrm{km\,s^{-1}\,kpc^{-1}}$ at the start of each run.

7 Tests

The scheme was checked with a line-by-line Python version of the code, using a 300-star Plummer sphere of $1000\,\mathrm{M}_\odot$ with a Plummer radius of $1\,\mathrm{pc}$, integrated for $8\,\mathrm{Myr}$ with $\eta = 0.02$:

- The analytic tidal jerk agrees with a finite-difference derivative of the acceleration along the motion, with an error that falls linearly with the difference step, as it should.
- With the solar tide, $|\Delta E_\mathrm{J}/E_\mathrm{J}| = 1.7 \times 10^{-7}$, comparable to the isolated cluster (5.6×10^{-7} with the standard predictor).
- With a tide 1000 times stronger than the solar value ($r_\mathrm{J} \simeq 1.4\,\mathrm{pc}$, so the cluster is strongly stripped), $|\Delta E_\mathrm{J}/E_\mathrm{J}| = 3.6\times10^{-6}$ with the higher-order predictor, compared with 2.5×10^{-3} with the standard predictor, at the same cost.

8 Parameter file

```
&tides
  tidal = 'solar' ! 'none', 'solar' or 'tensor'
  oorta = 15.3 ! Oort A (km/s/kpc)
  oortb = -11.9 ! Oort B (km/s/kpc)
  rho0 = 0.097 ! local total density (Msun/pc^3)
  ! used only when tidal = 'tensor', units (km/s/kpc)^2 and km/s/kpc
  tij(1,1) = 1664.6, tij(3,3) = -5427.5
  omegat = 27.2
/
```

References

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